

# MISO LOLE-LOLH Evaluation and Stress Testing

*A Chronological Resource Adequacy Model Built on 2021-2025 EIA-930 Data and the MISO PY 2026-2027 LOLE Study Report*

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## 1. Purpose

This project builds a portfolio-grade approximation of MISO's Loss of Load Expectation (LOLE) and Loss of Load Hours (LOLH) resource adequacy process. It uses five years (2021-2025) of real, hourly EIA-930 balancing-authority data for the MISO footprint, calibrated against figures published in MISO's Planning Year 2026-2027 LOLE Study Report, to:

- Establish a chronological (hour-by-hour) baseline reliability assessment against the industry-standard 0.1 day/year (1-day-in-10-years) criterion;
- Stress-test that baseline against two forward-looking risk drivers MISO itself has flagged in recent studies, rapid data center load growth and correlated winter freeze-out events; and
- Evaluate battery energy storage as a mitigation lever, subject to realistic power, energy, and efficiency constraints.

This model demonstrates the underlying methodology used in production resource adequacy studies: chronological net load construction, probabilistic capacity modeling, iterative reliability calibration, scenario-based stress testing, and constrained dispatch optimization, applied end to end using real data. It is a simplified approximation and does not replicate MISO's production SERVIM-based study.

## 2. Data & Methodology

### 2.1 Data Sources

| Source                                                               | Use                                                                                |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------|
| EIA-930 hourly BALANCE files (2021-2025, Balancing Authority = MISO) | Hourly demand, wind, and solar generation for the full MISO footprint              |
| MISO PY 2026-2027 LOLE Study Report                                  | Seasonal PRM, peak demand, system-wide weighted EFORD, cold-weather outage derates |

The raw EIA-930 files contain all ~65 U.S. balancing authorities per file; the ingestion pipeline filters to the MISO row (the RTO's full footprint, already aggregated, so no zonal reconstruction is required) and harmonizes across a schema change EIA-930 introduced mid-2024, where solar and wind generation columns were split into "with/without integrated battery storage" components. Both schemas are detected automatically by header inspection and reconciled into unified demand/solar/wind series.

### 2.2 Continuity & Imputation

Hourly data is reindexed onto a strictly continuous UTC timestamp range spanning the full study period, exposing any missing or duplicated hours (including daylight-saving-time artifacts) as NaN. Short gaps ( $\leq 6$  hours) are linearly interpolated; longer gaps fall back to the same hour one week prior. Local time (for season

and hour-of-day labeling) is derived via timezone conversion off the clean UTC index. This approach avoids the classic 23/25-hour DST bug entirely, since it does not depend on the raw local-time fields in the source data.

On the source dataset used here, this pipeline confirmed zero missing hours and zero NaNs across all 43,824 hours from January 1, 2021 through December 31, 2025. The imputation logic is exercised on every run and was verified to be a no-op on this data.

## 2.3 Net Load Construction

Hourly net load is computed as Demand – Wind Generation – Solar Generation, and segmented into MISO's four evaluation seasons (Winter: Dec-Feb, Spring: Mar-May, Summer: Jun-Aug, Fall: Sep-Nov).

## 2.4 Reliability Engine

The core of the model is a chronological Monte Carlo capacity-adequacy engine:

- Each season's thermal fleet is represented as a set of generic, class-average capacity blocks (400 MW each), sized to a target Unforced Capacity (UCAP) and de-rated by MISO's own published seasonal system-wide weighted EFORD (Summer 7.45%, Fall 8.93%, Winter 10.85%, Spring 9.29%, from Tables 3-1 through 3-4 of the LOLE Study Report). A small fixed adder represents firm non-thermal capacity (hydro, biomass, imports).
- For each calendar day in the five-year record, a Bernoulli outage draw is made per block (forced outages persist on an hour-to-day timescale, so day-level draws are the model's simplifying granularity) and held constant across that day's 24 hours.
- An hour is a loss-of-load (LOL) event whenever net load exceeds that day's available capacity. LOLE is the expected number of distinct LOL days per year; LOLH is the expected number of LOL hours per season; Expected Unserved Energy (EUE) is the expected shortfall magnitude (MWh/season).
- Capacity calibration: the model applies the classic PRM formula,  $UCAP = Peak \times (1 + PRM)$ , to each season's own historical peak net load, then runs a bisection search on a capacity scaling factor until simulated LOLE converges to the 0.1 day/year target. This keeps the calibration self-consistent with the data actually being simulated, since MISO's forward-looking PY2026-2027 forecast peak sits well above what 2021-2025 actually produced. The bisection search mirrors MISO's own described iterative process of adding or removing proxy capacity until the model reaches criterion (LOLE Study Report, Section 3.7.2).

## 3. Stress Test Scenarios

### 3.1 Scenario A: Data Center Load Surge

Models rapid, always-on industrial/data-center demand growth as two additive components layered on baseline net load:

- A flat, 6,000 MW block, ramped linearly from 0 MW in 2021 to full magnitude by 2025, representing the near-constant (roughly 90% or higher load factor) draw of hyperscale compute load. This load shape stresses shoulder and overnight hours that historically carried large reserve margins, in addition to the traditional afternoon peak.
- An additional 2,000 MW coincident adder applied only to each season's top-decile (90th-percentile) demand hours, representing cooling load that scales with weather-driven peaks.

### 3.2 Scenario B: Correlated Winter Freeze-Out

Models a MISO-reported emerging risk driver: thermal generator outages that correlate with extreme cold. Because EIA-930 contains no temperature field, cold-snap hours are proxied as the top 5% of winter demand hours (542 hours across the five-year record). Winter peak demand in MISO's footprint is heating-driven and strongly correlated with cold snaps. During those flagged hours:

- Wind output is derated by 55%, reflecting icing-related underperformance risk; and
- The winter thermal fleet's available capacity is reduced by 11,320 MW, MISO's own reported incremental cold-weather outage capacity for the winter season (LOLE Study Report, Table 1-1, row B.10).

## 4. Storage Mitigation Methodology

A 5,000 MW / 28,000 MWh multi-tranche battery fleet is dispatched as a peak-shaving mitigation against Scenario B, split into a 3,000 MW / 4-hour tranche (12,000 MWh) and a 2,000 MW / 8-hour tranche (16,000 MWh). This sizing falls within MISO's currently registered storage capacity range and the report's own multi-gigawatt battery fleet trajectory.

Dispatch is a chronological, greedy peak-shaving controller enforcing three physical constraints every hour:

- Power rating: charge and discharge cannot exceed the tranche's rated MW;
- State-of-charge bounds: energy stored is held within a 10-90% depth-of-discharge band of each tranche's energy capacity; and
- Round-trip efficiency (85%): applied as a  $\sqrt{0.85} \approx 92.2\%$  one-way efficiency on both the charge and discharge legs.

The controller discharges whenever net load is in the top decile of its season (the hours that actually drive LOLE risk) and charges whenever net load is in the bottom quartile, subject to the constraints above. Tranches are dispatched in sequence so the second tranche responds to the load shape already shaved by the first.

## 5. Results & Discussion

### 5.1 Seasonal LOLE Summary (days/year)

| Case                          | Summer | Fall  | Winter | Spring |
|-------------------------------|--------|-------|--------|--------|
| Baseline (calibrated)         | 0.110  | 0.107 | 0.111  | 0.114  |
| Scenario A: Data Center Surge | 1.039  | 1.369 | 1.441  | 0.612  |
| Scenario B: Winter Freeze-Out | 0.110  | 0.107 | 10.815 | 0.114  |
| Post-Storage Mitigation       | 0.109  | 0.107 | 9.837  | 0.114  |

*0.1 day/year is the reliability criterion (1-day-in-10-years).*

### 5.2 Baseline Calibration

The bisection-calibrated capacity fleet lands within roughly 0.01-0.014 days/year of the 0.1 target in every season, about as close as five discrete historical weather-years can resolve a criterion this small (see Section 6.1). Calibrated seasonal UCAP ranges from approximately 87.2 GW (Spring) to 113.4 GW (Summer), against historical peak net loads of 86.2 GW and 113.2 GW respectively.

### 5.3 Scenario A: Broad, Cross-Seasonal Strain

The data center load surge raises LOLE by roughly 6x to 13x in every season, including seasons beyond the one it might be assumed to target most. This is a direct consequence of the always-on load shape: because the added demand never goes to zero, it erodes reserve margin during hours that were historically low-risk (nights, shoulder seasons), which is precisely the concern driving MISO's recent scrutiny of large flexible-load interconnection requests.

### 5.4 Scenario B: Concentrated, Severe Winter Risk

The correlated freeze-out scenario raises Winter LOLE from 0.111 to 10.815 days/year, roughly a 100x increase, while leaving every other season unaffected, since the stress is winter-specific by construction. Winter Expected Unserved Energy rises from 403 MWh/season at baseline to over 418,600 MWh/season under stress. This result is consistent with MISO's own reported finding that correlated cold-weather-driven outages are now the fastest-growing risk driver in its footprint, and that annual risk has begun shifting from Summer toward Winter in recent planning cycles.

### 5.5 Storage Mitigation: Partial Relief, Clear Limits

Deploying the 5,000 MW / 28,000 MWh battery fleet against Scenario B reduces Winter LOLE from 10.815 to 9.837 days/year, an improvement of roughly 9%, leaving the system still nearly 100x above the 0.1 day/year criterion. The dispatch trace shows why: the fleet fully discharges within the first day of the multi-day cold event and does not recover, because net load stays elevated above the charging trigger for the remainder of the event. A short-duration battery fleet can shave the leading edge of a sustained, correlated stress event. It cannot carry a multi-day, 11+ GW capacity shortfall on its own, which underscores the need for longer-duration storage, demand response, or firm capacity as complements to short-duration battery fleets when addressing correlated extreme-weather risk.

## 6. Engineering Discussion

### 6.1 Sample-Size Granularity

With only five realized historical weather years, LOLE is a coarse, almost step-like function of capacity: one day flipping into or out of shortfall moves annualized LOLE by  $1/5 = 0.2$  days/year, twice the 0.1 day/year target itself. MISO's actual study resolves this using a 30+ year synthetic weather-year library; the calibration search here logs a warning whenever it cannot fully converge within tolerance for this reason. This is the primary source of residual noise in the baseline results and would shrink with a longer weather-year sample.

### 6.2 Model Simplifications

This is a simplified, system-wide approximation of MISO's production process. Notable simplifications, and why they don't invalidate the exercise:

- Single system-wide capacity pool: no Local Resource Zone (LRZ) or transmission-constrained modeling; MISO's real study layers 10 zonal Local Clearing Requirement analyses on top of the system-wide PRM analysis.
- Generic class-average thermal blocks with MISO's published system-wide weighted EFORD, in place of unit-level GADS outage records for every real generator.
- Day-level forced-outage persistence in the Monte Carlo draws, in place of sub-daily outage/repair event sequences.

- Cold-snap detection via a winter-demand percentile proxy, since EIA-930 has no temperature field. This proxy is directionally consistent with MISO's own load/weather regression approach.

### 6.3 Interpretation

These simplifications mean the model's absolute figures will not match MISO's official study numbers. Its methodology, chronological net load construction against a probabilistic capacity draw, iterative calibration to a defined reliability criterion, scenario-based stress testing, and constrained storage dispatch, mirrors the structure of the real process closely enough to support meaningful engineering conclusions about relative risk and mitigation effectiveness.

## 7. Conclusion

Under a simplified, real-data-grounded chronological LOLE/LOLH framework, the MISO footprint's baseline reliability calibrates close to its 0.1 day/year target. Two independent, MISO-relevant stress mechanisms, always-on data center load growth and correlated winter freeze-out events, each drive material reliability degradation, with the winter freeze-out scenario producing a two-order-of-magnitude increase in Winter LOLE. A multi-gigawatt battery fleet provides only partial, front-loaded relief against a sustained correlated event. Resource adequacy planning for these emerging risk drivers calls for a diversified toolkit: long-duration storage, demand response, and firm capacity. A single mitigation lever alone is not sufficient.

## Data Sources

- U.S. Energy Information Administration, EIA-930 Hourly Electric Grid Monitor, Balancing Authority data, 2021-2025.
- MISO, Planning Year 2026-2027 Loss of Load Expectation Study Report (Tables 1-1, 3-1 through 3-4).